Journal of Alloys and Compounds 589 (2014) 192–199

Contents lists available at ScienceDirect

Journal of Alloys and Compounds

journal homepage: www.elsevier.com/locate/jalcom

Review

Thermodynamic modeling of the RE–Pb (RE = Sc, Dy, Gd) systems

A. Iddaoudi [a,*], C. Servant [b], N. Selhaoui [a], S. Kardellass [a], K. Mahdouk [a], L. Bouirden [a]

[a] Laboratoire de Thermodynamique et Energétique, LTE, Université Ibn-Zohr, B.P. 8106 Agadir, Morocco
[b] Laboratoire de Physicochimie de l'Etat Solide, ICMMO, Université Paris-Sud, 91405 Orsay Cedex, France

ARTICLE INFO

Article history:
Received 26 August 2013
Received in revised form 19 November 2013
Accepted 20 November 2013
Available online 6 December 2013

Keywords:
Sc–Pb
Dy–Pb
Gd–Pb systems
Phase diagrams
CALPHAD method
Thermodynamic description

ABSTRACT

The phase diagrams and thermodynamic properties of RE–Pb (RE = Sc, Dy, Gd) systems have been assessed by means of the CALPHAD method. The solution phases (liquid, fcc, bcc and hcp) were described by the sublattice formalism and the excess term of the Gibbs energy with the Redlich–Kister equation. The stoichiometric intermetallic compounds (Sc_5Pb_3, Sc_6Pb_5, Dy_5Pb_4, DyPb, $DyPb_2$, $DyPb_3$, β-Gd_5Pb_4, α-Gd_5Pb_4, $Gd_{11}Pb_{10}$, Gd_6Pb_7, $GdPb_2$ and $GdPb_2$) were modeled as line compounds. The non-stoichiometric Dy_5Pb_3 and Gd_5Pb_3 phases with a narrow homogeneity range were modeled using a two-sublattice model with substitution. A consistent set of the thermodynamic parameters leading to a reasonable agreement between the calculated results and literature data was obtained.

© 2013 Elsevier B.V. All rights reserved.

Contents

1. Introduction . . . 192
2. Review of experimental data . . . 193
   2.1. The Sc–Pb system . . . 193
   2.2. The Dy–Pb system . . . 193
   2.3. The Gd–Pb system . . . 193
3. Thermodynamic models . . . 194
   3.1. Pure elements . . . 194
   3.2. Solution phases . . . 194
   3.3. Stoichiometric compounds . . . 194
   3.4. Intermediate phases . . . 194
4. Results and discussions . . . 194
   4.1. The Sc–Pb system . . . 196
   4.2. The Dy–Pb system . . . 196
   4.3. The Gd–Pb system . . . 197
5. Conclusions . . . 199
   Appendix A. Supplementary material . . . 199
   References . . . 199

1. Introduction

The alloy systems or intermetallic compounds constituted of rare earth (RE) and Pb are generally used as magnetic materials [1,2]. Thus, a complete thermodynamic description of the RE–Pb (RE = Sc, Dy, Gd) systems is needed. The present paper is a part of a thermodynamic description of RE–Pb (RE = Lu [3] and Yb [4]) already calculated in our laboratory which is required to give a better understanding of their functional properties as materials of technological interest of these alloys. The objective of this study is to develop a thermodynamic description of the RE–Pb (Re = Sc, Dy, Gd) binary systems by means of the CALPHAD [5] method.

* Corresponding author. Tel.: +212 671321888.
E-mail addresses: iddaoudi19@gmail.com (A. Iddaoudi), colette.servant@orange.fr (C. Servant).

0925-8388/$ - see front matter © 2013 Elsevier B.V. All rights reserved.
http://dx.doi.org/10.1016/j.jallcom.2013.11.131

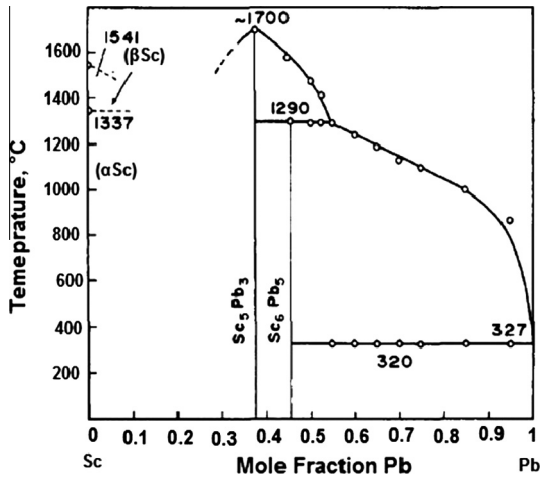

Fig. 1. The phase diagram of the Sc–Pb system reviewed by Palenzona and Manfrinetti [6].

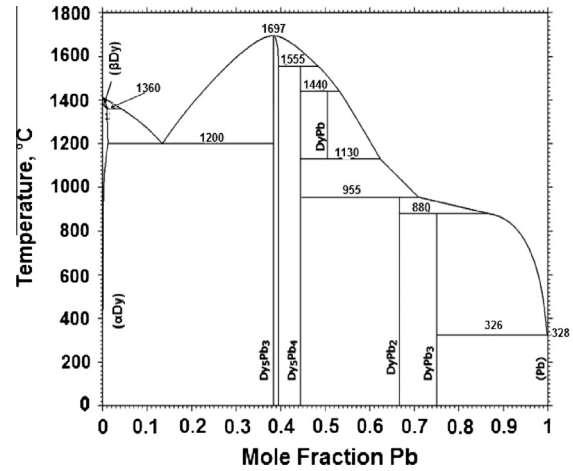

Fig. 2. The phase diagram of the Dy–Pb system reviewed by Massalski et al. [10].

2. Review of experimental data

2.1. The Sc–Pb system

The Sc–Pb phase diagram has been investigated by Palenzona and Manfrinetti [6] in the range 30–100 at.% Pb (see Fig. 1) using differential thermal analysis (DTA), metallographic analysis, X-ray diffraction (XRD) and electron microscopy. The Sc-rich part of this system has not been studied because of the high melting temperatures of the corresponding alloys and contamination of them by molybdenum which was used as container material during preparation of the samples. Two compounds were described: Sc_5Pb_3 and Sc_6Pb_5. For the Sc_5Pb_3 compound a congruent melting was determined at 1700 °C, while for the Sc_6Pb_5 compound a peritectic formation was reported, at 1290 °C. The crystal structures of the various phases are reported in Table 1.

No integral or partial enthalpies of mixing of the liquid phase were published in the literature. Shubin et al. [7], and Yamtshikov et al. [8] have measured the standard enthalpy of formation of Sc_6Pb_5 using the ElectroMotive Force (EMF). Shubin et al. [7] have measured the partial energy of Sc in the liquid alloys (L + Pb_5Sc_6).

2.2. The Dy–Pb system

The Dy–Pb system consists of three terminal solution phases (αDy, βDy and Pb), and five intermetallic compounds (Dy_5Pb_3, Dy_5Pb_4, DyPb, $DyPb_2$ and $DyPb_3$). The phase diagram of the Dy–Pb system was first investigated by McMasters et al. [9] using metallography, X-ray diffraction and differential thermal analysis. Five intermetallic compounds were described: Dy_5Pb_3, Dy_5Pb_4, DyPb, $DyPb_2$ and $DyPb_3$. For the Dy_5Pb_3 compound a congruent melting was determined at 1697 °C, while for the Dy_5Pb_4, DyPb, $DyPb_2$ and $DyPb_3$ compounds, a peritectic formation was reported, respectively, at 1555 °C, 1440 °C, 955 °C and 880 °C. Later the phase diagram of the Dy–Pb system has been reviewed by Massalski [10], it is shown in Fig. 2. The crystal structures of the various phases are reported in Table 1.

Colinet and Pasturel [11] calculated the enthalpies of mixing of the liquid phase at 50 at.% Pb and the limiting partial heat of solution in the Dy–Pb system. Palenzona and Cirafici [12], Yamshchikov et al. [13] and Bayanov et al. [14] have measured the standard enthalpy of formation of $DyPb_3$ using the dynamic differential calorimetry method and the ElectroMotive Force (EMF) respectively. Colinet and Pasturel [11] calculated the enthalpies of formation for five intermetallic compounds: Dy_5Pb_3, Dy_5Pb_4, DyPb, $DyPb_2$ and $DyPb_3$.

2.3. The Gd–Pb system

The Gd–Pb system consists of three terminal solution phases (αGd, βGd and Pb), and six intermetallic compounds (Gd_5Pb_3, Gd_5Pb_4, $Gd_{11}Pb_{10}$, Gd_6Pb_7, $GdPb_2$ and $GdPb_3$). Demel and Gschneidner [15] have first studied the Gd–Pb phase diagram by means of differential thermal analysis (DTA), optical metallography, and X-ray diffraction methods. Six intermediate phases were found (Gd_5Pb_3, Gd_5Pb_4, $Gd_{11}Pb_{10}$, Gd_6Pb_7, $GdPb_2$ and $GdPb_3$). The compounds Gd_5Pb_4, $Gd_{11}Pb_{10}$, Gd_6Pb_7, $GdPb_2$ and $GdPb_3$ are formed by peritectic reaction at 1460 °C, 1225 °C, 1160 °C, 1010 °C and 969 °C, respectively. The Gd_5Pb_3 phase melts congruently at 1670 °C. A

Table 1
Symbols and crystal structures of the stable solid phases in the Sc–Pb, Dy–Pb and Gd–Pb systems.

System	Diagram symbol	Composition at.% Pb	Pearson symbol	Symbol used in thermo-calc data file	Prototype
Sc–Pb	(β-Sc)	0	*cI2*	BCC_A2	W
	(α-Sc)	0	*hP2*	HCP_A3	Mg
	Sc_5Pb_3	37.5	*hP16*	Sc_5Pb_3	Mn_5Si_3
	Sc_6Pb_5	45.5	*oI44*	Sc_6Pb_5	Ti_6Ge_5
	(Pb)	100	*cF4*	FCC_A1	Cu
Dy–Pb	(β-Dy)	0	*cI2*	BCC_A2	W
	(α-Dy)	0	*hP2*	HCP_A3	Mg
	Dy_5Pb_3	37.5	*hP16*	Dy_5Pb_3	Mn_5Si_3
	Dy_5Pb_4	44.4	*oP36*	Dy_5Pb_4	Sm_5Ge_4
	DyPb	50	–	DyPb	–
	$DyPb_2$	66.7	–	$DyPb_2$	–
	$DyPb_3$	75	*cP4*	$DyPb_3$	Cu_3Au
	(Pb)	100	*cF4*	FCC_A1	Cu
Gd–Pb	(β-Gd)	0	*cI2*	BCC_A2	W
	(α-Gd)	0	*hP2*	HCP_A3	Mg
	Gd_5Pb_3	37.5	*hP16*	Gd_5Pb_3	Mn_5Si_3
	β-Gd_5Pb_4	44.4	–	Gd_5Pb_4_HT	–
	α-Gd_5Pb_4	44.4	*oP36*	Gd_5Pb_4_LT	Sm_5Ge_4
	$Gd_{11}Pb_{10}$	47.6	–	$Gd_{11}Pb_{10}$	–
	Gd_6Pb_7	53.8	–	Gd_6Pb_7	–
	$GdPb_2$	66.7	–	$GdPb_2$	–
	$GdPb_3$	75	*cP4*	$GdPb_3$	Cu_3Au
	(Pb)	100	*cF4*	FCC_A1	Cu

polymorphic transition ($\alpha \leftrightarrow \beta$) was determined at 1410 °C for Gd_5Pb_4. The Gd–Pb system has been reviewed by Palenzona and Cirafici [16], it is plotted in Fig. 3. The crystal structures of the various phases are reported in Table 1.

Colinet and Pasturel [11] calculated the enthalpies of mixing of the liquid phase at 50 at.% Pb and the limiting partial heat of solution in the Gd–Pb system. Palenzona and Cirafici [12], Roshchina and Bayanov [17] have measured the standard enthalpy of formation of $GdPb_3$ using the dynamic differential calorimetry and the ElectroMotive Force (EMF) methods respectively. Colinet and Pasturel [11] calculated the enthalpies of formation for four intermetallic compounds: Gd_5Pb_3, Gd_5Pb_4, $GdPb_2$ and $GdPb_3$.

3. Thermodynamic models

3.1. Pure elements

The Gibbs energy of the pure element i (i = Sc, Dy, Gd, Pb) in the phase φ (φ = Liquid, HCP_A3, BCC_A2 and FCC_A1), referred to the enthalpy of its stable state at 298.15 K, is described as a function of temperature by

$$\begin{aligned} {}^0G_i^{\varphi} &= G_i^{\varphi}(T) - H_i^{SER}(298.15\ \mathrm{K}) \\ &= a + bT + cT\ \ln T + dT^2 + eT^3 + fT^7 + gT^{-1} + hT^{-9} \end{aligned} \quad (1)$$

where $H_i^{SER}(298.15\ \mathrm{K})$ is the molar enthalpy of the element i at 298.15 K in its Standard Element Reference (SER) state, HCP_A3 for Sc, Dy, Gd and FCC_A1 for Pb.

In this paper, the Gibbs energy functions are taken from the SGTE compilation of Dinsdale [18].

3.2. Solution phases

In the RE–Pb (RE = Sc, Dy, Gd) binary (A–B) systems, the solution phases, (φ = Liquid, BCC_A2 and HCP_A3) were modeled as substitutional solutions according to the polynomial Redlich–Kister model [19]. The molar Gibbs free energies of the solution phase are given as follows:

$$G_m^{\varphi} = \sum_{i=A,B} {}^0G_i^{\varphi} x_i + RT \sum_{i=A,B} x_i \ln x_i + x_A x_B \sum_{j=0}^{n} {}^jL_{A,B}^{\varphi}(x_A - x_B)^j \quad (2)$$

where ${}^0G_i^{\varphi}$ is the Gibbs free energy of the pure component i (i = Sc, Dy, Gd, Pb) in the reference state with φ phase, x_i represents the mole fraction of the component i, R is the gas constant, T the temperature in Kelvin, The binary interaction parameters of the ${}^jL_{A,B}^{\varphi}$ type assessed in this study were temperature dependent as follows:

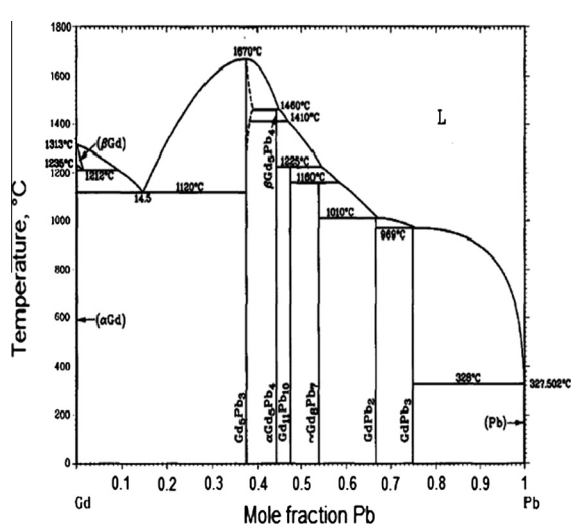

Fig. 3. The phase diagram of the Gd–Pb system reviewed by Palenzona and Cirafici [16].

$${}^jL_{A,B}^{\varphi} = a_j + b_j T \quad (3)$$

The interaction coefficients a_j and b_j (j = 0, 1, 2,...) are to be evaluated in the optimization process.

3.3. Stoichiometric compounds

The Gibbs energy of the stoichiometric compounds (Sc_5Pb_3, Sc_6Pb_5, Dy_5Pb_4, DyPb, $DyPb_2$, $DyPb_3$, α-Gd_5Pb_4, β-Gd_5Pb_4, $Gd_{11}Pb_{10}$, Gd_6Pb_7, $GdPb_2$ and $GdPb_3$) noted as ${}^0G_{A_pB_q}$ was expressed as follows:

$${}^0G_{A_pB_q} = \frac{p}{p+q}{}^0G_A + \frac{q}{p+q}{}^0G_B + a + bT \quad (4)$$

where ${}^0G_A$ and ${}^0G_B$ are the Gibbs energy of the pure elements Pb and i (i = Sc, Dy, Gd,), respectively, a and b are parameters to be determined.

3.4. Intermediate phases

The intermetallic compounds of the Gd_5Pb_3 and Dy_5Pb_3 phases (substoichiometric in Re) in the Re–Pb (Re: Gd and Dy) binary systems have an experimental small homogeneity range, they were treated by a two-sublattice model as $(Re)_{0.625}(Re,Pb\%)_{0.375}$ The symbol % denotes the major component in the considered sublattice. The Gibbs energy function per mole (m) of the formula unit $(Re)_{0.625}(Re,Pb\%)_{0.375}$ is the following:

$$\begin{aligned} G^{0.625Re0.375Pb} - H_{0.625Re0.375Pb}^{SER} = {} & y_{Re}^1 y_{Pb}^2\, {}^0G^{Re:Pb} + y_{Re}^1 y_{Re}^2\, {}^0G^{Re:Re} \\ & + RT[0.625 y_{Re}^1 \ln y_{Re}^1 \\ & + 0.375(y_{Re}^2 \ln y_{Re}^2 + y_{Pb}^2 \ln y_{Pb}^2)] \\ & + {}^{ex}G_m \end{aligned} \quad (5)$$

$$H_{0.625Re0.375Pb}^{SER} = 0.625 H_{Re}^{SER} + 0.375 H_{Pb}^{SER} \quad (6)$$

where $y_{Re}^1 = 1$ denotes the site fraction of Re in the first sublattice, y_{Re}^2 and y_{Pb}^2 the site fractions of Re and Pb in the second sublattice, $G^{0.625Re0.375Pb}$ is the Gibbs energy of the hypothetical compound 0.625Re0.375Re.

${}^{ex}G_m$ is the excess Gibbs energy expressed by the following expression:

$${}^{ex}G_m = y_{Re}^2 y_{Pb}^2 (L_{Re,Pb:Pb}^{\lambda}) \quad (7)$$

where $L_{Re,Pb:Pb}^{\lambda}$ represents the interaction parameters between the elements Re and Pb in the first sublattice while the second sublattice is only occupied by the element Re.

These excess parameters are temperature dependent as:

$$L_{Re,Pb:Pb}^{\lambda} = a_{\lambda} + b_{\lambda} T \quad (8)$$

In order to avoid the occurrence of the hypothetical compound 0.625Re0.375Re during the phase diagram calculation, the value +5000 J/mol of atoms was added to the function GHSERPb, see Tables 3 and 4.

4. Results and discussions

All of the experimental data mentioned in Section 2 were used for the evaluation of the thermodynamic model parameters. The optimization of the thermodynamic parameters in the RE–Pb (RE = Sc, Dy, Gd) systems was performed with the PARROT [20] module of the Thermo-calc software [21]. The thermodynamic parameters of all phases in these three systems were evaluated and they are respectively listed in Tables 2–4.

Table 2
The optimized thermodynamic parameters of the Sc–Pb system.

Phase	Thermodynamic models	Parameters in SI
Liquid	$(Sc,Pb)_1$	${}^0L^{liq} = -77,737 + -7.378T$ ${}^1L^{liq} = 42,557 - 10.653T$ ${}^2L^{liq} = -17,110 + 29.971T$
HCP_A3	$(Sc,Pb)_1(Va)_{0.5}$	${}^0L^{HCP_A3} = -41,899 - 5.872T$
BCC_A2	$(Sc,Pb)_1(Va)_3$	${}^0L^{BCC_A2} = -68,365 + 4.022T$
FCC_A1	(Pb)	No excess term
Sc_5Pb_3	$(Sc)_{0.625}:(Pb)_{0.375}$	$G^{Sc5Pb3}_{Sc:Pb} - 0.625^{298}H^{HCP_A3}_{Sc} - 0.375^{298}H^{FCC_A1}_{Pb} = 0.625G^{HCP_A3}_{Sc} + 0.375G^{FCC_A1}_{Pb} - 40,352 + 0.873T$
Sc_6Pb_5	$(Sc)_{0.545}:(Pb)_{0.455}$	$G^{Sc6Pb5}_{Sc:Pb} - 0.545^{298}H^{HCP_A3}_{Sc} - 0.455H^{FCC_A1}_{Pb} = 0.545G^{HCP_A3}_{Sc} + 0.455G^{FCC_A1}_{Pb} - 49,397 + 8.142T$

Table 3
The optimized thermodynamic parameters of the Dy–Pb system.

Phase	Thermodynamic models	Parameters in SI
Liquid	$(Dy,Pb)_1$	${}^0L^{liq} = -110,331 + 2.933T$ ${}^1L^{liq} = 1536 - 0.374T$ ${}^2L^{liq} = -19,434 + 29.794T$
HCP_A3	$(Dy,Pb)_1(Va)_{0.5}$	${}^0L^{HCP_A3} = 43,546 - 64.298T$
BCC_A2	$(Dy,Pb)_1(Va)_3$	${}^0L^{BCC-A2} = -50,413 - 11.393T$
FCC_A1	(Pb)	No excess term
Dy_5Pb_3	$(Dy)_{0.625}:(Dy,Pb)_{0.375}$	$G^{Dy5Pb3}_{Dy:Dy} - {}^{298}H^{HCP_A3}_{Dy} = G^{HCP_A3}_{Dy} + 5000G^{Dy5Pb3}_{Dy:Pb} - 0.625^{298}H^{HCP_A3}_{Dy} - 0.375^{298}H^{FCC_A1}_{Pb} = 0.625G^{HCP_A3}_{Dy} + 0.375G^{FCC_A1}_{Pb} - 43,686 + 1.622T$ ${}^0L^{Dy5Pb3}_{Dy,Pb:Pb} = -17,387 + 3.801T$ ${}^1L^{Dy5Pb3}_{Dy,Pb:Pb} = -7839$
Dy_5Pb_4	$(Dy)_{0.556}:(Pb)_{0.444}$	$G^{Dy5Pb4}_{Dy:Pb} - 0.556^{298}H^{HCP_A3}_{Dy} - 0.444H^{FCC_A1}_{Pb} = 0.556G^{HCP_A3}_{Dy} + 0.556G^{FCC_A1}_{Pb} - 45,192 + 1.874T$
DyPb	$(Dy)_{0.500}:(Pb)_{0.500}$	$G^{DyPb}_{Dy:Pb} - 0.500^{298}H^{HCP_A3}_{Dy} - 0.500H^{FCC_A1}_{Pb} = 0.500G^{HCP_A3}_{Dy} + 0.500G^{FCC_A1}_{Pb} - 35,826 - 3.272T$
$DyPb_2$	$(Dy)_{0.333}:(Pb)_{0.667}$	$G^{DyPb2}_{Dy:Pb} - 0.333^{298}H^{HCP_A3}_{Dy} - 0.667H^{FCC_A1}_{Pb} = 0.333G^{HCP_A3}_{Dy} + 0.667G^{FCC_A1}_{Pb} - 44,416 + 9.883T$
$DyPb_3$	$(Dy)_{0.250}:(Pb)_{0.750}$	$G^{DyPb3}_{Dy:Pb} - 0.25^{298}H^{HCP_A3}_{Dy} - 0.75H^{FCC_A1}_{Pb} = 0.25G^{HCP_A3}_{Dy} + 0.75G^{FCC_A1}_{Pb} - 39,039 + 10.388T$

Table 4
The optimized thermodynamic parameters of the Gd–Pb system.

Phase	Thermodynamic models	Parameters in SI
Liquid	$(Gd,Pb)_1$	${}^0L^{liq} = -136233 + 2.561T$ ${}^1L^{liq} = -35,487 + 33.406T$ ${}^2L^{liq} = 36,790 - 2.576T$
HCP_A3	$(Gd,Pb)_1(Va)_{0.5}$	${}^0L^{HCP_A3} = 15,169 - 61.892T$
BCC_A2	$(Gd,Pb)_1(Va)_3$	${}^0L^{BCC-A2} = -65,441 - 11.393T$ ${}^1L^{BCC-A2} = 975$
FCC_A1	(Pb)	No excess term
Gd_5Pb_3	$(Gd,Pb)_{0.625}:(Pb)_{0.375}$	$G^{Gd5Pb3}_{Gd:Gd} - {}^{298}H^{HCP_A3}_{Gd} = G^{HCP_A3}_{Gd} + 5000G^{Gd5Pb3}_{Gd:Pb} - 0.625^{298}H^{HCP_A3}_{Gd} - 0.375^{298}H^{FCC_A1}_{Pb} = 0.625G^{HCP_A3}_{Gd} + 0.375G^{FCC_A1}_{Pb} - 48,055 + 1.480T$ ${}^0L^{Gd5Pb3}_{Gd,Pb:Pb} = -30 + 2.999T$ ${}^1L^{Gd5Pb3}_{Gd,Pb:Pb} = -39,844$
Gd_5Pb_4_LT	$(Gd)_{0.556}:(Pb)_{0.444}$	$G^{Gd5Pb4_LT}_{Gd:Pb} - 0.556^{298}H^{HCP_A3}_{Gd} - 0.444H^{FCC_A1}_{Pb} = 0.556G^{HCP_A3}_{Gd} + 0.556G^{FCC_A1}_{Pb} - 54,244 + 4.112T$
Gd_5Pb_4_HT	$(Gd)_{0.556}:(Pb)_{0.444}$	$G^{Gd5Pb4_HT}_{Gd:Pb} - 0.556^{298}H^{HCP_A3}_{Gd} - 0.444H^{FCC_A1}_{Pb} = 0.556G^{HCP_A3}_{Gd} + 0.556G^{FCC_A1}_{Pb} - 54,037 + 3.989T$
$Gd_{11}Pb_{10}$	$(Gd)_{0.524}:(Pb)_{0.476}$	$G^{Gd11Pb10}_{Gd:Pb} - 0.524^{298}H^{HCP_A3}_{Gd} - 0.476H^{FCC_A1}_{Pb} = 0.524G^{HCP_A3}_{Gd} + 0.476G^{FCC_A1}_{Pb} - 57,164 + 6.528T$
Gd_6Pb_7	$(Gd)_{0.462}:(Pb)_{0.538}$	$G^{Gd6Pb7}_{Gd:Pb} - 0.462^{298}H^{HCP_A3}_{Gd} - 0.538H^{FCC_A1}_{Pb} = 0.462G^{HCP_A3}_{Gd} + 0.538G^{FCC_A1}_{Pb} - 62,889 + 11.883T$
$GdPb_2$	$(Gd)_{0.333}:(Pb)_{0.667}$	$G^{GdPb2}_{Gd:Pb} - 0.333^{298}H^{HCP_A3}_{Gd} - 0.667H^{FCC_A1}_{Pb} = 0.333G^{HCP_A3}_{Gd} + 0.667G^{FCC_A1}_{Pb} - 47,563 + 6.437T$
$GdPb_3$	$(Gd)_{0.250}:(Pb)_{0.750}$	$G^{GdPb3}_{Gd:Pb} - 0.25^{298}H^{HCP_A3}_{Gd} - 0.75H^{FCC_A1}_{Pb} = 0.25G^{HCP_A3}_{Gd} + 0.75G^{FCC_A1}_{Pb} - 35,201 + 1.824T$

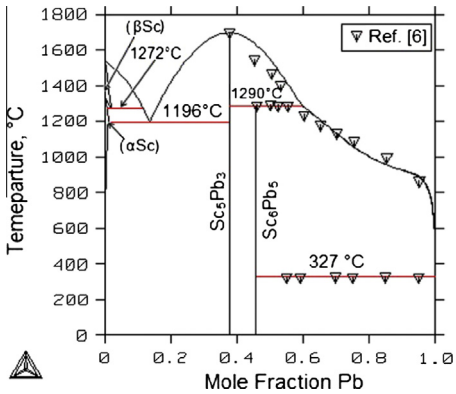

Fig. 4. Calculated Sc–Pb phase diagram compared with the experimental data [6].

4.1. The Sc–Pb system

The Sc–Pb calculated phase diagram compared with the experimental data is shown in Fig. 4. A good agreement is noted between the calculated phase diagram and the experimental data.

The calculated invariant reactions are listed in Table 5 and compared with the experimental results from [6]. For the Sc-rich part of this system we suggested the eutectic reaction of Liq ↔ Sc_5Pb_3 + (αSc) at 1196 °C and the catatectic reaction Liq + αSc) ↔ (βSc) at 1272 °C according to the literature data of Dy-rich part of Dy–Pb system and presented in Fig. 5. The calculated enthalpies of formation of the intermetallic compounds are compared with the experimental data and plotted in Fig. 6. The calculated enthalpies agree well with the experimental data [7,8].

Fig. 7 presents the partial energy of Sc in the liquid alloys (L + Pb_5Sc_6) in temperature range of 610–1140 K compared with the ones measured by Shubin et al. [7]. A large difference is noted with the experimental data.

4.2. The Dy–Pb system

The Dy–Pb calculated phase diagram is presented in Fig. 8. The calculated compositions and temperatures for the invariant reactions are listed in Table 5 and compared with the experimental results from [10]. The Dy-rich part of this system is plotted in Fig. 9. The calculated results are in reasonable agreement with most of the experimental data [10].

Table 5
Calculated invariant reactions compared with the literature data in the Sc–Pb, Dy–Pb and Gd–Pb systems.

Systems	Invariant reactions	*T* (°C)	Composition (at.% Pb)			References
Dy–Pb	Liq ↔ Pb + $DyPb_3$	326	99.7	100	75	[10]
		325	99.7	100	75	This work
	Liq + $DyPb_3$ ↔ $DyPb_2$	880	86	75	66.7	[10]
		863	80	75	66.7	This work
	Liq + $DyPb_2$ ↔ Dy_5Pb_4	955	71	66.7	44.4	[10]
		963	70	66.7	44.4	This work
	Liq + DyPb ↔ Dy_5Pb_4	1130	62	50	44.4	[10]
		1130	66	50	44.4	This work
	Liq + DyPb ↔ Dy_5Pb_4	1440	53	50	44.4	[10]
		1447	55	50	44.4	This work
	Liq + Dy_5Pb_4 ↔ Dy_5Pb_3	1555	48	44.4	39	[10]
		1553	49	44.4	38	This work
	Liq ↔ Dy_5Pb_3	1695	–	37.5	–	[10]
		1690	–	37.5	–	This work
	Liq ↔ Dy_5Pb_3 + (αDy)	1200	13.5	37.5	1.2	[10]
		1996	12.3	37.5	0.65	This work
	Liq + (αDy) ↔ (βDy)	1360	4	1.2	0.5	[10]
		1357	4	0.97	0.5	This work
Gd–Pb	Liq ↔ Pb + $GdPb_3$	328	99.7	100	75	[15]
		327	100	100	75	This work
	Liq + $GdPb_3$ ↔ $GdPb_2$	969	76	75	66.7	[15]
		968	75.77	75	66.7	This work
	Liq + $GdPb_2$ ↔ Gd_6Pb_7	1010	68	66.7	53.8	[15]
		1015	70.87	66.7	53.8	This work
	Liq + Gd_6Pb_7 ↔ $Gd_{11}Pb_{10}$	1160	59	53.8	47.6	[15]
		1164	62.04	53.8	47.6	This work
	Liq + $Gd_{11}Pb_{10}$ ↔ α-Gd_5Pb_4	1225	56	47.6	44.4	[15]
		1226	59.61	47.6	44.4	This work
	α-Gd_5Pb_4 ↔ β-Gd_5Pb_4	1410	–	44.4	–	[15]
		1410	–	44.4	–	This work
	Liq + β-Gd_5Pb_4 ↔ Gd_5Pb_3	1460	45	44.4	37.5	[15]
		1461	49.85	44.4	37.5	This work
	Liq ↔ Gd_5Pb_3	1670	37.5	37.5	–	[15]
		1668	37.5	37.5	–	This work
	Liq ↔ Gd_5Pb_3 + (αGd)	1120	14.5	37.5	0.5	[15]
		1114	12.84	37.5	0.8	This work
	Liq + (αGd) ↔ (βGd)	1212	8	2	1	[15]
		1202	8	1.5	1	This work
Sc–Pb	Liq ↔ Pb + Sc_6Pb_5	320	99.7	100	45.5	[6]
		327	99.9	100	45.5	This work
	Liq + Sc_6Pb_5 ↔ Sc_5Pb_3	1290	55	45.5	37.5	[6]
		1290	59	45.5	37.5	This work
	Liq ↔ Sc_5Pb_3	1700	–	37.5	–	[6]
		1697	–	37.5	–	This work
	Liq ↔ Sc_5Pb_3 + (αSc)	1196	13.5	37.5	1	This work
	Liq + (αSc) ↔ (βSc)	1272	11	1.8	0	This work

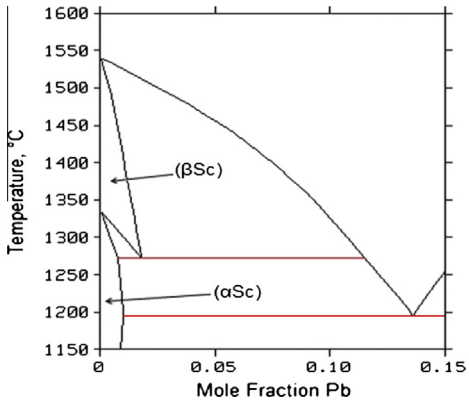

Fig. 5. Sc-rich region of the Sc–Pb phase diagram.

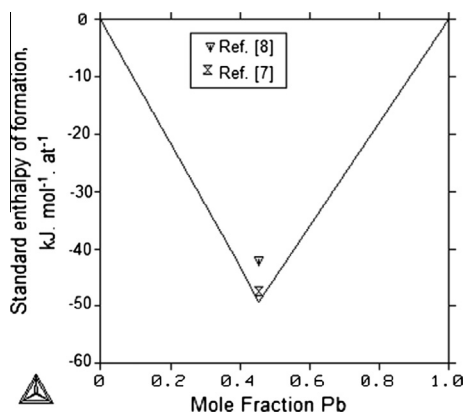

Fig. 6. Calculated standard enthalpies of formation of the intermetallic compounds in the Sc–Pb system compared with the experimental data [7,8]. The reference states are HCP_A3 (Sc) and FCC_A1 (Pb) at $T = 298$ K.

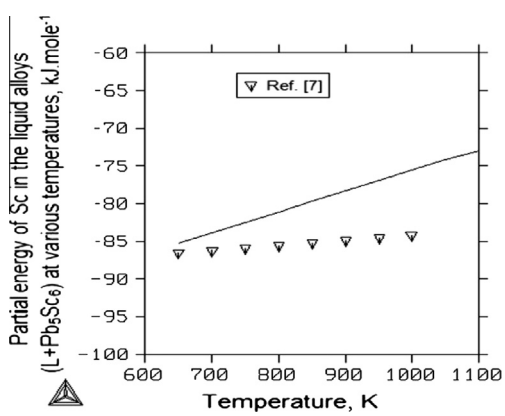

Fig. 7. Calculated partial energy of Sc in the liquid alloys ($L + Pb_5Sc_6$) at various temperatures compared with the experimental data [7].

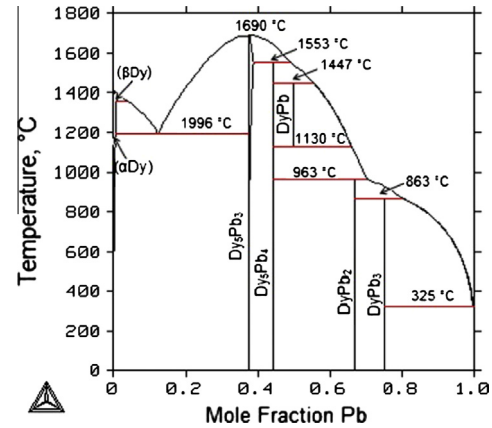

Fig. 8. Calculated Dy–Pb phase diagram.

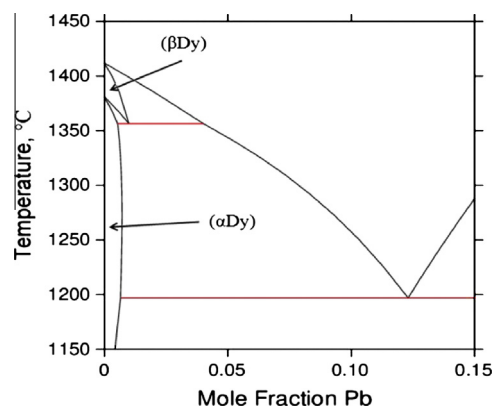

Fig. 9. Dy-rich region of the Dy–Pb phase diagram.

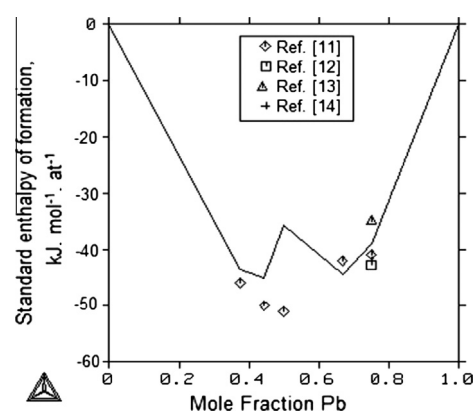

Fig. 10. Calculated standard enthalpies of formation of the intermetallic compounds in the Dy–Pb system compared with the experimental data [11–14]. The reference states are HCP_A3 (Dy) and FCC_A1 (Pb) at $T = 298$ K.

The calculated enthalpies of formation of the intermetallic compounds are compared with experimental data and plotted in Fig. 10. The calculated enthalpies agree well with the experimental data [11–14] except the calculated enthalpy of formation of DyPb ($\Delta H_f = -35.82$ kJ/mol) which is lower than the predicted one by Colinet and Pasturel [11] ($\Delta H_f = -51$ kJ/mol).

Fig. 11 shows the calculated enthalpies of mixing of Dysprosium and Lead of the liquid phase compared with the ones predicted by Ref. [11]. The maximum is noted at 50 at.% Pb and −27.58 kJ/mol. A very good agreement is noted.

The calculated partial enthalpies of mixing of Dysprosium and Lead of the liquid phase compared with the ones predicted by Ref. [11] are plotted in Fig. 12. A satisfactory agreement is observed with the predicted values in Dy-rich and Pb-rich regions.

4.3. The Gd–Pb system

Fig. 13 shows the calculated equilibrium phase diagram with all the experimental data used in our optimization. The Gd-rich part of this system is plotted in Fig. 14. A satisfactory agreement is noted

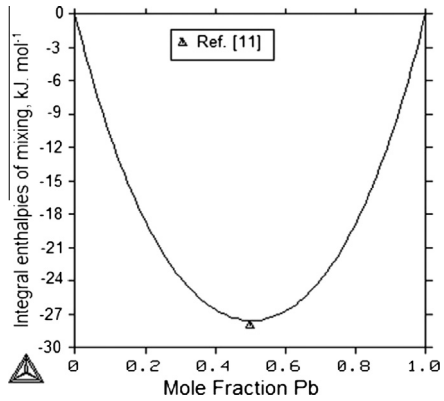

Fig. 11. Calculated enthalpies of mixing of liquid phase compared with the predicted data [11]. The reference states are liquid (Dy) and liquid (Pb) at $T = 2000$ K.

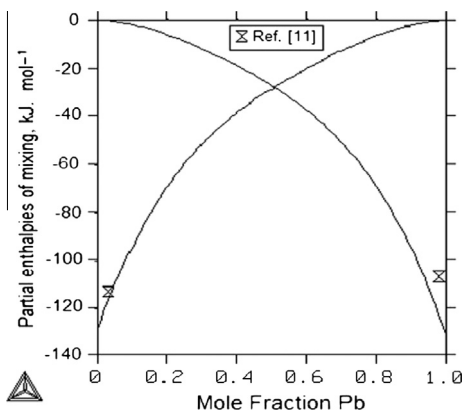

Fig. 12. Calculated partial enthalpies of mixing of liquid phase compared with the predicted data [11]. The reference states are liquid (Dy) and liquid (Pb) at $T = 2000$ K.

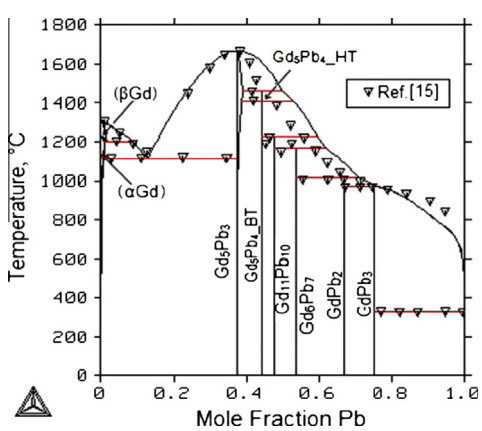

Fig. 13. Calculated Gd–Pb phase diagram compared with the experimental data [15].

with most of the experimental data. The calculated compositions and temperatures for the invariant reactions are listed in Table 5 and compared with the experimental results from [15]. The calculated results are in reasonable agreement with most of the experimental data [15].

The calculated enthalpies of formation of the intermetallic compounds compared with the experimental data are plotted in Fig. 15. The calculated enthalpies agree well with the experimental by data [11,12,17].

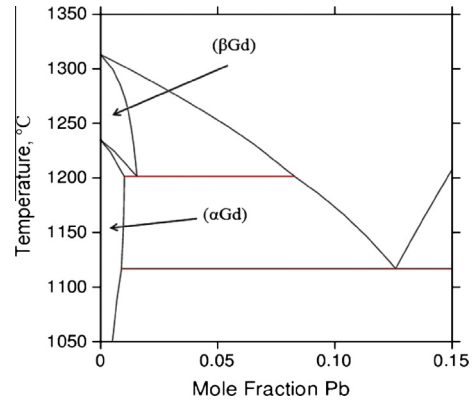

Fig. 14. Gd-rich region of the Gd–Pb phase diagram.

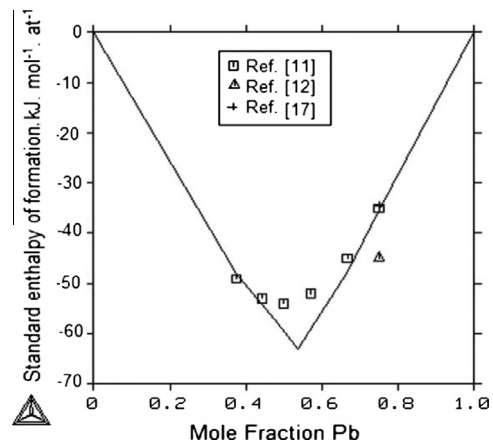

Fig. 15. Calculated standard enthalpies of formation of the intermetallic compounds in the Gd–Pb system compared with the experimental data [11,12,17]. The reference states are HCP_A3 (Gd) and FCC_A1 (Pb) at $T = 298$ K.

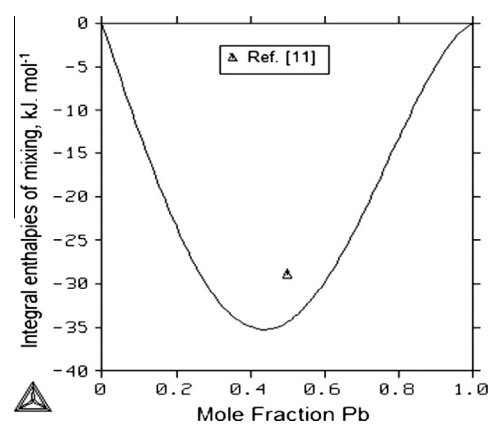

Fig. 16. Calculated enthalpies of mixing of liquid phase compared with the predicted data [11]. The reference states are liquid (Gd) and liquid (Pb) at $T = 2000$ K.

As mentioned in [22], we confirmed that the enthalpies of formation of the polymorphous are very similar. In order to calculate this small difference in the structural energies of the polytypes, and the right transition temperature Gd_5Pb_4_LT ↔ Gd_5Pb_4_HT, it is necessary to use all the digits of the optimized thermodynamic parameters in Table 4.

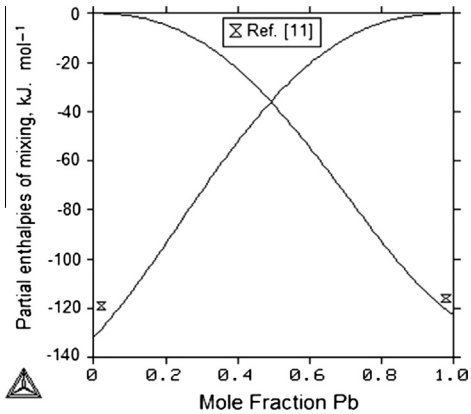

Fig. 17. Calculated partial enthalpies of mixing of liquid phase compared with the predicted data [11]. The reference states are liquid (Gd) and liquid (Pb) at $T = 2000$ K.

Fig. 16 shows the calculated enthalpies of mixing of Gadolinium and Lead of the liquid phase compared with the ones predicted by Colinet and Pasturel [11]. The maximum is noted at 50 at.% Pb and −33.51 kJ/mol. A satisfactory agreement is noted.

The calculated partial enthalpies of mixing of Gadolinium and Lead of the liquid phase compared with the ones predicted by Ref. [11] are plotted in Fig. 17. A satisfactory agreement is observed with the predicted values in the Gd-rich and Pb-rich regions.

The phase diagrams RE–Pb (RE = Sc, Dy, Gd) become more complex when the addition of Pb goes from Sc, to Dy then to Gd. All diagrams present the RE_5Pb_3 intermetallic compound with a congruent melting.

5. Conclusions

The phase diagrams and thermodynamic properties of RE–Pb (RE = Sc, Dy, Gd) binary systems have been critically assessed from the experimental information available in the literature. A self-consistent set of thermodynamic parameters has been evaluated by optimization. It can faithfully reproduce most of the experimental information for the RE–Pb (RE = Sc, Dy, Gd) systems.

Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.jallcom.2013.11.131.

References

[1] R.D. Hutchens, W.E. Wallace, J. Solid State Chem. 3 (1971) 564.
[2] B. Stalinski, AIP Conf. Proc. 18 (1974) 490.
[3] A. Iddaoudi, N. Selhaoui, M. Ait Amar, K. Mahdouk, A. Aharoune, L. Bouirden, J. Therm. Anal. Calorim. 110 (2012) 923.
[4] M. Idbenali, C. Servant, N. Selhaoui, L. Bouirden, Calphad 33 (2009) 570.
[5] L. Kaufman, H. Bernstein, Computer Calculation of Phase Diagram, Academic Press, New York, 1970.
[6] A. Palenzona, P. Manfrinetti, J. Alloys Comp. 220 (1995) 157.
[7] A. Shubin, K. Shunyaev, L. Yamshchikov, Arch. Metall. Mater. 53 (2008) 1119.
[8] L.F. Yamshchikov, V.A. Lebedev, S.P. Raspopin, S.E. Puchinskis, Russ. J. Phys. Chem. 59 (12) (1985) 1757.
[9] O.D. McMasters, T.J. O'Keefe, K.A. Gschneidner, Trans. Metall. Soc. AIME 242 (1968) 936.
[10] T.B. Massalski, H. Okamoto, P.R. Subramanian, L. Kacprzak, Binary Alloy Phase Diagrams, second ed., ASM, Materials Park, OH, USA, 1990.
[11] C. Colinet, A. Pasturel, A. Percheron-Guegan, J.C. Achard, J. Less Common Met. 102 (1984) 239.
[12] A. Palenzona, S. Cirafici, Thermochim. Acta 6 (1973) 55.
[13] L.E. Yamshchikov, A.V. Volynchuk, S.V. Kurochkin, E.N. Sattarov, S.R. Raspopin, Izv. Akad. Nauk SSSR, Met. 3 (1989) 204 (in Russian);
L.E. Yamshchikov, A.V. Volynchuk, S.V. Kurochkin, E.N. Sattarov, S.R. Raspopin, TR: Russ. Metall. 3 (1989) 195.
[14] A.P. Bayanov, E.N. Ganchenko, Yu.A. Afanasev, Zh. Fiz. Khim. 51 (1977) 216 (in Russian);
A.P. Bayanov, E.N. Ganchenko, Yu.A. Afanasev, TR: Russ. J. Phys. Chem. 51 (1977) 122.
[15] J.T. Demel, K.A. Gschneidner Jr., J. Nucl. Mater. 29 (1969) 111–120.
[16] A. Palenzona, S. Ciraflci, J. Phase Equilib. 12 (1991) 686.
[17] V.R. Roshchina, A.P. Bayanov, Izv. Akad. Nauk SSSR, Met. 4 (1974) 183 (in Russian);
V.R. Roshchina, A.P. Bayanov, TR: Russ. Metall. 4 (1974) 131.
[18] A.T. Dinsdale, Calphad 15 (1991) 317.
[19] O. Redlich, A. Kister, Ind. Eng. Chem. 40 (1948) 345.
[20] B. Jansson, Thesis, Royal Institute of Technology, Stockholm, 1984.
[21] B. Sundman, J.-O. Andersson, Calphad 9 (1985) 153.
[22] F. Stein, M. Palm, G. Sauthoff, Intermetallics 13 (2005) 1056.